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Department of Computer Science and Engineering

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Course: B.Tech

Semester: - 3rd

Branch: CSE/CS/AI/CPS/DS

Subject Code: - 3003

Max. Marks:60

Subject Name: Computer Organization and Architecture

Time: 03hr.

NOTE:

1. All the questions are compulsory.
2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.
4. Answers without precision receive no credit.

1.A	A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. Compute the average memory access time in ns. ?	5
✓ B.	Consider 64MB main memory and 16KB cache divided into 128 bytes/block. Compute size of Physical address, no. of TAG bits, TAG comparators for the following mappings (Assume word size is one byte): (a) Direct mapping (b) Associate mapping (c) 8-way Set-associate mapping	5
✓ 2.A	Assume you are tasked with designing a hardware architecture for a cache memory mapping technique. To accomplish this, you are first required to analyze and perform a trade-off between different memory mapping techniques in terms of their hardware requirements and conflict issues. Based on your analysis, identify the mapping technique you prefer and design its hardware architecture. You may assume appropriate configurations for main memory and cache memory according to your design.	5
B.	Consider a disk having 256 tracks, 512 sectors per track, 2048 bytes/ sector. The disk is rotating with 3000rpm. Seek time is 10msec. Compute the followings: (a) Capacity of disk and number of bits required to address the disk. (b) What is the data transfer rate, If the disk is used in cycle stealing mode of DMA such that whenever 64-bit word is available, it will be transferred in 16 μ s? What is the % of time processor is blocked due to DMA?	2+3

P.T.O.

3.A A 32KB Direct mapped cache is organized into 128-byte blocks. A two-dimensional Array A 512x512 is manipulated by two programs P1 and P2, given below. Each element of the array occupies 8 bytes and array stored in ROW-major order. What will be the total number of miss references (if represented using M1 and M2 respectively) with respect to the programs P1 and P2.

Program P1-->

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        X+=A[i][j];
    }
}
```

Program P2-->

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        X+=A[j][i];
    }
}
```

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(a) A 5-stage pipeline (IF, ID, EX, MEM, WB) has an ideal CPI = 1. Instruction mix: 60% instructions cause no stall, 25% cause 1 data-hazard stall, 15% cause 2 control-hazard stalls. An equivalent unpipelined processor completes one instruction in 5 clock cycles. Calculate: (i) the pipeline average CPI, (ii) speedup of pipeline vs unpipelined assuming same clock period.

✓ (b) Consider, a hypothetical 4-stage pipeline in which different instructions spent different number of clocks. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below. What is the number of cycles needed to execute them as per the following loop?

2+3

For (i= 1 to 2) {I1;I2;I3;I4}.

	S1	S2	S3	S4
I1	1	3	2	1
I2	2	1	2	2
I3	1	1	1	1
I4	2	1	3	1

4.A. An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (Ex), memory access (MEM), and register write back (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively. To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 2.2/3 ns. Also, the Ex stage is split into two stages (Ex1, Ex2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the Ex stage and produce the next instruction pointer at the end of the Ex stage in the old design and at the end of the Ex2 stage in the new design. The IF stage stalls after fetching a

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P.T.O

	branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is __?	
	Show how the following floating-point arithmetic are performed using IEEE 754 single-precision rules (normalize, align exponents, add mantissas, normalize, round), (where significands are truncated to 4 decimal digits). (a) $5.566 \times 10^{-2} + 7.777 \times 10^{-3}$ (b) $8.844 \times 10^{-2} - 2.233 \times 10^{-1}$ B. (or) A processor of 50MHz require 1500 clocks for initializing and terminating the DMA. It is required to transfer 400KB file from 2000KBps device. What is the percentage of processor time is used in this DMA?	5
5.A	(a) Trade-off between Program driven and Interrupt driven data transfer approaches, in terms of CPU utilization. (b) Design the architecture of Horizontal microprogrammed control unit. Analyze the time required for control signal generation	2+3
B.	With a neat schematic, discuss all the necessary steps involved in transferring data using Direct Memory Access (DMA) controller with proper instruction/signal flow . Also, compare the performance under its different operating modes (based on when the system bus is returned during the DMA transfer).	5
6.A	(a) Assume a processor uses microprogrammed control. Write the sequence of microoperations that occur when an interrupt is acknowledged and control is transferred to the ISR. Clearly indicate how the return-from-interrupt is handled. (b) Consider a <u>Vertical</u> Microprogramming control unit (CU), in which the system has to support 128-instructions. Each instruction on avg. requires 16μ-op. The system is using 63 control signals and 16 Flag conditions have to be supported by this CU. Further, <u>one address control instruction</u> is used for control word sequencing. (i) What is the size of control word? (ii) How much space (in Kilobytes) required for control memory?	2+3
B.	(a) Analyze the number of 1-Address and 2-Address instruction that may be required to perform (Clearly mention the number of instructions). $X = (M+N*O) / (P*Q)$	2+3

(b) Consider a processor which allow only register operands in the instructions. Each instruction can have at most two source operands and one destination operand. All variables are dead after the following code segment

```
c = a + b;  
d = c * a;  
e = c + a;  
x = c * c;  
If (x > a)  
{ y = a * a; }  
else  
{ d = d * d;  
  e = e * e;  
}
```

- (1) if Spilling is not allowed and code motion is the only one that is allowed for Optimization. What will be the minimum number of registers required for this?
- (2) if spilling is allowed, then compute the number of memory accesses required during the code execution?